GIT Histology & Embryology

(General Histological features of GIT
Differentiate among the parts of small & large intestine on the basis of histology

Development of GIT
List the derivatives of foregut, midgut & hindgut)

1. The wall of the general gastrointestinal tract from the esophagus to the anal canal is organized into four distinct histological layers. What is the correct sequence of these layers from the internal lumen outward?
a. Serosa, Muscularis externa, Submucosa, Mucosa
b. Submucosa, Mucosa, Muscularis externa, Adventitia
c. Mucosa, Submucosa, Muscularis externa, and Serosa/Adventitia
d. Muscularis externa, Submucosa, Mucosa, Serosa
e. Mucosa, Muscularis externa, Submucosa, Adventitia
2. The innermost layer, the mucosa, is further subdivided into three micro-anatomical strata. The specific stratum composed of a thin sheet of smooth muscle fibers that drives localized folding of the mucous membrane is the:
a. Lamina propria
b. Epithelium
c. Submucosa
d. Muscularis mucosae
e. Adventitia proper
3. The lamina propria of the GIT mucosa is a supportive layer situated immediately beneath the surface epithelium. Histologically, this layer consists primarily of:
a. Dense compact bone trabeculae
b. Stratified squamous keratinized cords
c. Loose connective tissue rich in blood vessels, lymphatics, and diffuse lymphoid tissue (MALT)
d. Pure hyaline cartilage blocks
e. Multinucleated skeletal muscle cells
4. The muscularis externa layer throughout most of the gastrointestinal tract is composed of smooth muscle organized into two highly distinct patterns. These are the:
a. Inner longitudinal and outer circular layers
b. Oblique and transverse patterns
c. Inner circular and outer longitudinal layers
d. Vertical and diagonal quadrants
e. Random chaotic unmapped bundles
5. Located within the connective tissue layer of the submucosa is an autonomic nerve network that regulates mucosal movements and local glandular secretions. This is the Meissner's plexus, also known as the:
a. Myenteric plexus
b. Submucosal plexus
c. Celiac ganglion plexus

d. Hypogastric plexus
e. Cardiac plexus

6. Situated between the inner circular and outer longitudinal smooth muscle layers of the muscularis externa is a second autonomic network that controls peristaltic motility. This is the Auerbach's plexus, formally termed the:

a. Submucosal plexus
b. Deep mucosal plexus
c. Solar plexus
d. Myenteric plexus
e. Meissner's plexus

7. The small intestine is histologically adapted for massive nutrient absorption. Its lining displays permanent crescentic folds of mucosa and submucosa that extend micro-anatomically around the lumen, called the:

a. Rugae
b. Plicae circulares (valves of Kerckring)
c. Crypts of Lieberkuhn
d. Villi
e. Taeniae coli

8. Projecting directly from the mucosal surface of the small intestine into the lumen are millions of microscopic, finger-like extensions lined by simple columnar epithelium. These are the:

a. Microvilli
b. Villi
c. Plicae
d. Haustra
e. Crypts

9. Embedded within the core of loose connective tissue inside each individual intestinal villus is a specialized, blind-ended lymphatic capillary essential for fat absorption, known as a:

a. Sinusoid
b. Fenestrated venule
c. Lacteal
d. Crypt
e. Meissner canal

10. Simple tubular glands open into the spaces between the bases of the intestinal villi, extending down toward the muscularis mucosae. These glands are the:

a. Brunner's glands
b. Crypts of Lieberkühn (intestinal glands)
c. Gastric pits
d. Peyer's patches
e. Von Ebner glands

11. Microscopic examination of the simple columnar epithelium of the small intestine reveals cells with a prominent striated border on their apical surface. Under electron microscopy, this border is composed of thousands of:

a. Cilia
b. Stereocilia
c. Microvilli
d. Flagella

e. Pectinate ridges

12. Scattered abundantly among the absorptive enterocytes are pale, mucus-secreting cells that lubricate the intestinal walls. These are the:

a. Chief cells
b. Goblet cells
c. Paneth cells
d. Parietal cells
e. G-cells

13. Located selectively at the absolute base of the crypts of Lieberkühn are groups of exocrine cells featuring highly eosinophilic apical secretory granules that release lysozyme and defensins to destroy bacteria. These are the:

a. Enteroendocrine cells
b. Stem cells
c. Goblet cells
d. Paneth cells
e. M-cells

14. A definitive diagnostic histological feature of the duodenum that distinguishes it from all other segments of the small intestine is the presence of extensive branched tubuloalveolar mucous glands within its submucosa. These are the:

a. Crypts of Lieberkühn
b. Brunner's glands (duodenal glands)
c. Peyer's patches
d. Meibomian glands
e. Glands of Littre

15. Brunner's glands secrete a highly alkaline, bicarbonate-rich mucus into the duodenal lumen. The primary physiological role of this secretion is to:

a. Synthesize vitamin K
b. Activate pepsinogen into pepsin
c. Neutralize the acidic chyme entering from the stomach and protect the mucosa
d. Emulsify large fat globules
e. Stimulate skeletal muscle contractions

16. A key diagnostic feature of the jejunum under the microscope is that compared to the duodenum and ileum, it completely lacks submucosal Brunner's glands and features:

a. Zero villi or enterocytes
b. Tall, well-developed plicae circulares and long, prominent villi with an absence of Peyer's patches
c. Dense aggregates of encapsulated lymph nodes matching large intestine structures
d. A stratified squamous lining
e. Skeletal muscle in its muscularis externa

17. The ileum displays a highly specific histological marker. Its lamina propria and submucosa contain massive, unencapsulated structural aggregates of permanent lymphoid follicles known as:

a. Brunner's glands
b. Crypts of Lieberkühn
c. Peyer's patches
d. Tonsillar crypts
e. Hassall's corpuscles

18. Specialized epithelial cells called microfold cells overlay the lymphoid aggregates of Peyer's patches, functioning to sample and transport antigens from the lumen to underlying immune cells. These are:
a. Paneth cells
b. M-cells
c. Goblet cells
d. Enterocytes
e. Argentaffin cells
19. When comparing the small intestine to the large intestine under the microscope, the most striking structural difference characterizing the large intestine mucosa is the absolute absence of:
a. Villi and plicae circulares
b. Crypts of Lieberkühn
c. Goblet cells
d. Simple columnar cells
e. Lamina propria layers
20. Because the primary role of the large intestine is water absorption and fecal lubrication rather than nutrient digestion, its mucosal crypts are exceptionally deep and lined by an immense concentration of:
a. Paneth cells
b. Chief cells
c. Goblet cells
d. Parietal cells
e. Keratinized plates
21. In the cecum and colon, the outer longitudinal smooth muscle layer of the muscularis externa is modified. Instead of a uniform wrap, it condenses into three distinct, narrow ribbon-like bands visible macroscopically as the:
a. Haustra
b. Epiploic appendages
c. Taeniae coli
d. Plicae semilunares
e. Rugae bands
22. Tonic contractions of the taeniae coli bunch up the wall of the colon into a series of sacculations or pouches known macroscopically as:
a. Plicae circulares
b. Haustra (colonic sacculations)
c. Appendices epiploicae
d. Crypts of Lieberkühn
e. Valleculae
23. Projecting from the serosal surface of the large intestine colon are small, fat-filled pouches of peritoneum unique to this segment, classified as:
a. Taeniae coli
b. Plicae semilunares
c. Appendices epiploicae (omental appendages)
d. Meckel's diverticula
e. Solitary nodules

24. The vermiform appendix features an internal structure similar to the colon, but lacks taeniae coli. Its most prominent histological hallmark is a continuous, ring-like infiltration of the lamina propria and submucosa by:
a. Mucous Brunner's glands
b. Eosinophilic Paneth cells
c. Lymphoid follicles (forming a complete defensive ring)
d. Skeletal muscle bundles
e. Calcified sand grains
25. As the rectum transitions into the anal canal at the pectinate line, the simple columnar epithelial lining of the gastrointestinal tract undergoes a dramatic histological shift into:
a. Simple squamous epithelium
b. Stratified squamous non-keratinized epithelium
c. Pseudostratified ciliated columnar epithelium
d. Transitional epithelium
e. Stratified cuboidal blocks
26. The internal anal sphincter, which provides involuntary control over defecation, is formed by a localized thickening of which muscular layer?
a. Outer longitudinal smooth muscle layer of muscularis externa
b. Inner circular smooth muscle layer of the muscularis externa
c. Muscularis mucosae layer
d. Skeletal muscle of the pelvic floor diaphragm
e. Superficial fascia sheet
27. The external anal sphincter is under voluntary somatic control. Morphologically, it is composed of which muscle variety supplied by the pudendal nerve?
a. Smooth muscle fibers
b. Cardiac muscle branches
c. Skeletal muscle fibers
d. Myoepithelial cell nets
e. Elastic connective fibers
28. Let's shift completely to the embryological development of the CVS/GIT. During embryonic folding, the dorsal-most segment of the endoderm-lined yolk sac is incorporated into the embryo body to form the primitive digestive tube, known as the:
a. Notochord
b. Laryngotracheal diverticulum
c. Primitive gut tube
d. Vitelline duct
e. Allantois stalk
29. The primitive gut tube is divided into three distinct segments based on its arterial blood supply. The cranial-most segment, extending from the pharyngeal membrane down to the anterior portal entry of the bile duct, is the:
a. Midgut
b. Hindgut
c. Foregut
d. Tuberculum impar
e. Rathke pouch

30. The foregut undergoes extensive development. Which high-pressure systemic blood vessel branches off the abdominal aorta to supply all abdominal derivatives of the foregut?
a. Superior mesenteric artery
b. Inferior mesenteric artery
c. Celiac trunk (celiac artery)
d. Internal iliac artery
e. Renal artery trunk
31. Which of the following organs or structures develops as a direct embryological derivative of the foregut?
a. Cecum
b. Jejunum
c. Pharynx, esophagus, stomach, first/second parts of duodenum, liver, gallbladder, and pancreas
d. Descending colon
e. Urinary bladder
32. The stomach begins its development during the fourth week as a fusiform dilation of the foregut. This primitive dilation undergoes two distinct rotations. Its first rotation occurs around a vertical axis, spinning:
a. 90 degrees clockwise (causing the left side to face anteriorly and the right side to face posteriorly)
b. 180 degrees counter-clockwise
c. 45 degrees obliquely backward
d. 90 degrees counter-clockwise
e. Upward toward the thoracic inlet
33. Because the right vagus nerve blocks the posterior surface of the stomach following this 90-degree clockwise vertical rotation, it sets up the adult layout where the right vagus forms the:
a. Anterior vagal trunk
b. Posterior vagal trunk
c. Phrenic nerve connection
d. Recurrent laryngeal loop
e. Lingual nerve pathway
34. The liver and gallbladder develop from an endodermal outgrowth that buds off the ventral aspect of the distal foregut during the fourth week, known as the:
a. Rathke's pouch
b. Hepatic diverticulum (liver bud)
c. Pancreatic bud
d. Tuberculum impar
e. Vitelline tract
35. The pancreas develops from the fusion of two separate endodermal outgrowths arising from the duodenal lining. These structures are the:
a. Primary and secondary septa
b. Ventral and dorsal pancreatic buds
c. Hepatic and cystic channels
d. Alar and basal plates
e. Thymic and parathyroid pouches

36. During the rotation of the duodenum, the small ventral pancreatic bud migrates posteriorly around the gut tube to fuse with the larger dorsal pancreatic bud. The ventral pancreatic bud gives rise to the adult:
a. Tail of the pancreas
b. Body of the pancreas
c. Uncinate process and the lower part of the head of the pancreas
d. Main pancreatic duct accessory loop only
e. Minor palatine extensions
37. The midgut represents the intermediate segment of the primitive gut tube. It extends from the insertion point of the bile duct down to the junction of the:
a. Stomach and duodenum
b. Ileum and cecum
c. Proximal two-thirds and distal one-third of the transverse colon
d. Descending colon and sigmoid colon
e. Rectum and anal canal
38. The midgut derivatives are supplied with arterial blood by which major branch of the abdominal aorta?
a. Celiac trunk
b. Superior mesenteric artery
c. Inferior mesenteric artery
d. Internal thoracic artery
e. Common iliac artery
39. Which of the following structures is an embryological derivative originating from the midgut?
a. Esophagus
b. Stomach body
c. Jejunum, ileum, cecum, appendix, ascending colon, and proximal two-thirds of the transverse colon
d. Sigmoid colon
e. Upper half of the anal canal
40. Because of the rapid elongation of the midgut loop and the massive expansion of the fetal liver, the abdominal cavity becomes temporarily too small. During the sixth week, the midgut loops project out of the umbilicus into the extraembryonic coelom, a process termed:
a. Congenital omphalocele
b. Intestinal atresia
c. Physiological umbilical herniation
d. Meckel's diverticulation
e. Malrotation shifts
41. While residing inside the umbilical cord matrix, the primary midgut loop undergoes an initial counter-clockwise rotation of how many degrees around the axis of the superior mesenteric artery?
a. 90 degrees
b. 180 degrees
c. 270 degrees
d. 45 degrees
e. 360 degrees

42. During the tenth week of development, the herniated midgut loops return from the umbilical cord back into the expanded abdominal cavity. During this return, they execute a further counter-clockwise rotation of:
a. 90 degrees
b. 180 degrees (completing a total total counter-clockwise rotation of 270 degrees)
c. 360 degrees
d. 45 degrees
e. Zero degrees
43. If the midgut loops fail to return to the abdominal cavity from the umbilical cord postnatally, the infant is born with a midline defect where the abdominal viscera herniate through the umbilical ring covered by a translucent amniotic membrane. This birth defect is:
a. Gastroschisis
b. An omphalocele
c. Meckel's diverticulum
d. Congenital umbilical hernia
e. Hirschsprung's disease
44. A common congenital anomaly of the midgut occurs when a small remnant of the embryonic vitelline duct (yolk stalk) fails to obliterate, leaving a blind outpocketing on the antimesenteric border of the ileum. This structure is:
a. Zenker's diverticulum
b. A branchial cyst
c. Meckel's diverticulum (ileal diverticulum)
d. An omphalocele pouch
e. A thyroglossal tract pocket
45. Meckel's diverticulum is a high-yield clinical entity often remembered by the "rule of 2s." This rule specifies that the anomaly occurs in roughly 2% of the population, is situated approximately 2 feet proximal to the ileocecal valve, and averages how many inches in length?
a. 5 inches
b. 2 inches
c. 10 inches
d. 1 inch
e. 0.5 inches
46. The hindgut represents the final segment of the primitive gut tube. It extends from the distal one-third of the transverse colon down to the upper segment of the anal canal at the:
a. Ileocecal fold
b. Pectinate line
c. Sternal angle level
d. Killian's dehiscence
e. Linea alba line
47. All anatomical derivatives of the hindgut receive their primary systemic arterial blood supply via the:
a. Celiac trunk
b. Superior mesenteric artery
c. Inferior mesenteric artery
d. Internal iliac artery branches

e. Lumbar arteries

48. Which of the following structural components is derived directly from the embryonic hindgut?

a. Ascending colon
b. Cecum body
c. Distal one-third of the transverse colon, descending colon, sigmoid colon, rectum, and the upper part of the anal canal
d. Pancreas head
e. Gallbladder fossa

49. The terminal end of the hindgut expands into an endoderm-lined cavity called the cloaca. The cloaca is divided into an anterior urogenital sinus and a posterior anorectal canal by a downward sheet of mesoderm named the:

a. Tracheoesophageal septum
b. Urorectal septum
c. Septum transversum
d. Interatrial septum
e. Palatal shelf

50. Hirschsprung's disease (congenital megacolon) is a severe birth defect characterized by a localized absence of the myenteric (Auerbach's) and submucosal (Meissner's) nerve plexuses in a segment of the distal colon. This condition arises due to a failure of migration of:

a. Endodermal lining sheets
b. Neural crest cells into the hindgut wall during development
c. Splanchnic mesoderm layers
d. Ventral pancreatic bud loops
e. Left horn sinus venosus tracts

Answer Key Table (GIT Histology & Development -

Q—----------

1-8	9-16	17-24	25-32	33-40	41-48	49-50
c	c	c	b	b	c	b
d	b	b	c	b	b	b
c	d	a	c	b	b	
c	b	b	b	c	c	
b	c	c	c	b	b	
d	b	c	b	c	c	
b	c	c	c	c	c	
b	c	d	c	a	c	